Fundamentals of Electric Circuits

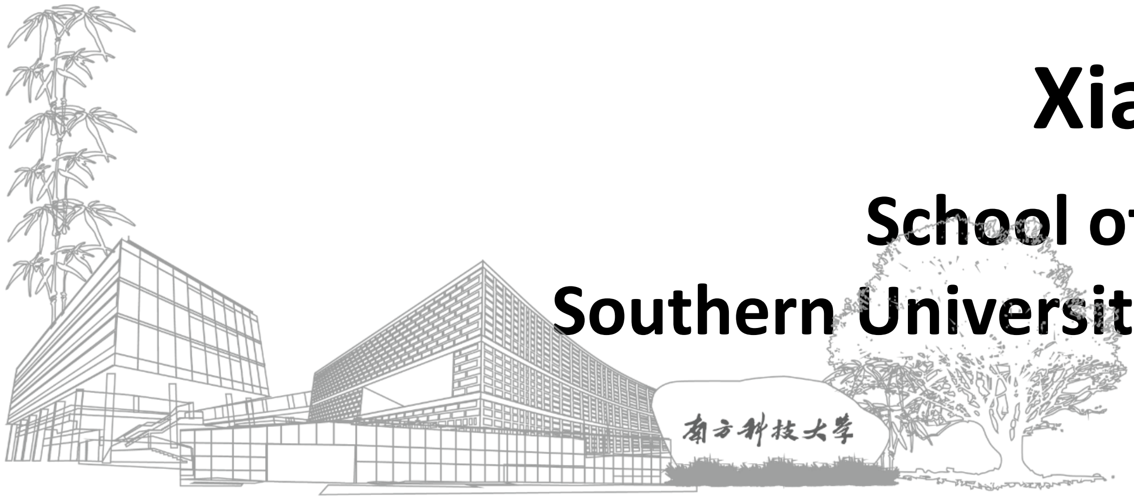
电路基础

Lecture3: Methods of Analysis

Xiaolong Liu

School of Microelectronics

Southern University of Science and Technology



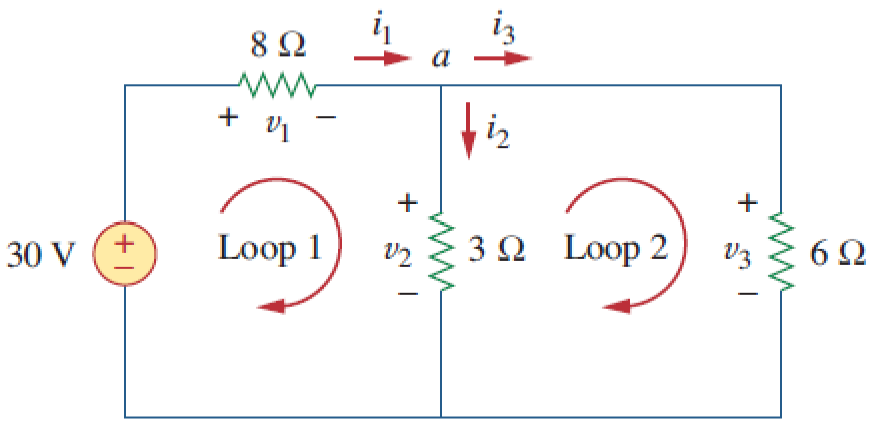
Outline

- Introduction
- Nodal Voltage Analysis (节点电压分析法)
- Nodal Analysis with Voltage Source
- Mesh Current Analysis (网孔电流分析法)
- Mesh Analysis with Current Source
- Nodal Versus Mesh Analysis
- DC Transistors Circuits

Introduction (I): Ohm's Law, KCL, KVL

Example 2.8 Find currents and voltages in the circuit

- Choose i_1, i_2, i_3, v_1, v_2, v_3 as unknown variables



$$\begin{cases} i_1 - i_2 - i_3 = 0 \\ -30 + v_1 + v_2 = 0 \\ -v_2 + v3 = 0 \\ v_1 = 8i_1 \\ v_2 = 3i_2 \\ v_3 = 6i_3 \end{cases}$$

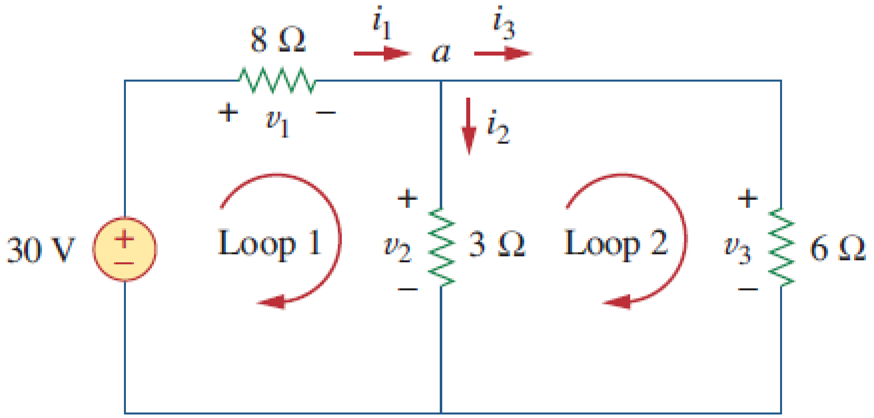
i_1=3A, i_2=2A, i_3=1A
v_1=24V, v_2=6V, v_3=6V

- One needs to solve six-dimensional linear equations

Introduction (II)

Example 2.8 Find currents and voltages in the circuit

- Choose loop currents i_1, i_3 as unknown variables



$$\begin{cases} -30 + 8i_1 + 3\times(i_1 - i_3) = 0 \\ 6i_3 + 3\times(i_3 - i_1) = 0 \end{cases}$$

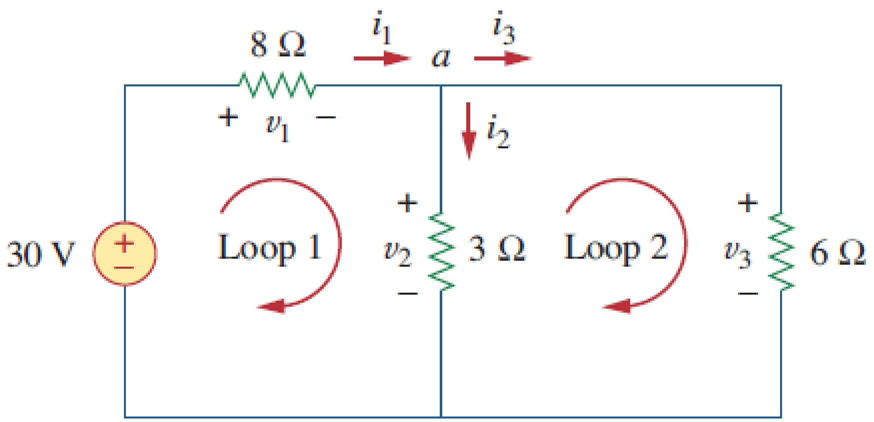
i_1=3A, i_3=1A → i_2=2A, v_1=24V, v_2=6V, v_3=6V

- One needs to solve two-dimensional linear equations

Introduction (III)

Example 2.8 Find currents and voltages in the circuit

- Choose the voltage at node *a*, v_2, as the only unknown variable



$$\frac{30-v_2}{8}-\frac{v_2}{3}-\frac{v_2}{6}=0$$

v_2=6V → i_1=3A, i_3=1A, i_2=2A, v_1=24V, v_3=6V

- The selection of unknown variables is important since it will determine the dimensionality of equations one need to solve

Outline

- Introduction
- **Nodal Voltage Analysis (节点电压分析法)**
- **Nodal Analysis with Voltage Source**
- Mesh Current Analysis (网孔电流分析法)
- Mesh Analysis with Current Source
- Nodal Versus Mesh Analysis
- DC Transistors Circuits

Nodal Voltage Analysis (节点电压分析法)

- Definition: Node voltages are selected as the independent variables for the description of the status of an electric circuit
  - KCL is used for writing equations

The steps (when there are no voltage sources):

1. Choose a node as the **reference node (参考节点)**, whose voltage default to zero (also called **ground**, ⏚). Label the voltages of the remaining n-1 nodes as $v_1, v_2, \ldots v_{n-1}$.
2. Apply **KCL** to each of the n-1 non-reference nodes. The branch currents are represented in terms of the node voltages according to the Ohm's law
3. Solve for the n-1 voltages from the resulting n-1 simultaneous equations
4. Derive any other voltages or currents from the node voltages if necessary

Example 1: Nodal Analysis

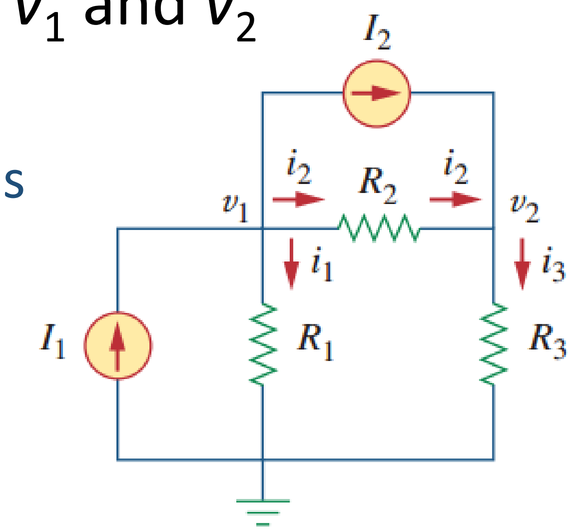
- Choose a node as the reference node
- Apply KCL law, representing the current i_1, i_2, i_3 in terms of v_1 and v_2

At node 1: $I_1 = I_2 + \dfrac{v_1}{R_1} + \dfrac{v_1 - v_2}{R_2}$

At node 2: $I_2 + \dfrac{v_1 - v_2}{R_2} = \dfrac{v_2}{R_3}$

In terms of the conductances

$$I_1 = I_2 + G_1 v_1 + G_2(v_1 - v_2)$$
$$I_2 + G_2(v_1 - v_2) = G_3 v_2$$



- Solve the simultaneous equations in the matrix form

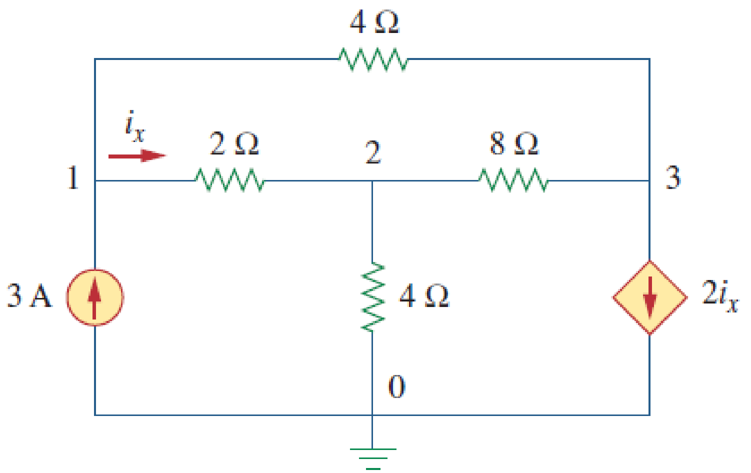
$$\begin{bmatrix} G_1 + G_2 & -G_2 \\ -G_2 & G_2 + G_3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} I_1 - I_2 \\ I_2 \end{bmatrix}$$

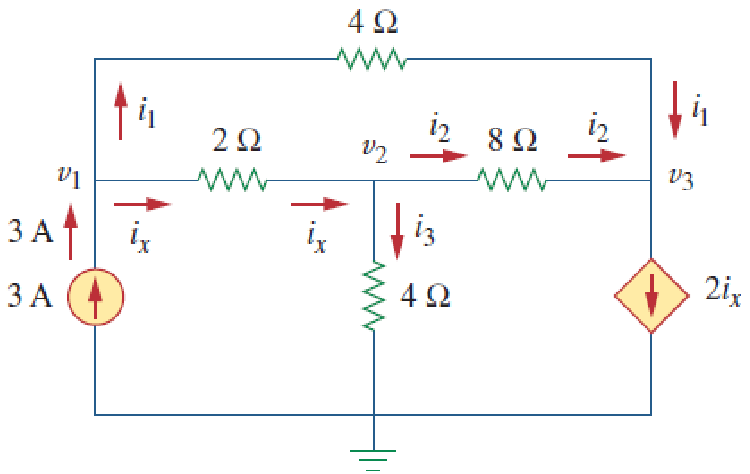
Q: Write KCL for reference node?

- The last step is to solve the system of equations, using substitution method, elimination method, Cramer's rule (克拉默法则)

Example 2: Nodal Analysis

Example 3.2 Determine the voltages at the nodes in Fig. 3.5(a).





At node 1: $3 = i_1 + i_x \quad \Rightarrow \quad 3 = \dfrac{v_1 - v_3}{4} + \dfrac{v_1 - v_2}{2}$

At node 2: $i_x = i_2 + i_3 \quad \Rightarrow \quad \dfrac{v_1 - v_2}{2} = \dfrac{v_2 - v_3}{8} + \dfrac{v_2 - 0}{4}$

At node 3: $i_1 + i_2 = 2i_x \quad \Rightarrow \quad \dfrac{v_1 - v_3}{4} + \dfrac{v_2 - v_3}{8} = \dfrac{2(v_1 - v_2)}{2}$

$$\begin{cases} 3v_1 - 2v_2 - v_3 = 12 \\ -4v_1 + 7v_2 - v_3 = 0 \\ 2v_1 - 3v_2 + v_3 = 0 \end{cases} \Rightarrow \quad \begin{aligned} v_1 &= 4.8\text{V} \\ v_2 &= 2.4\text{V} \\ v_3 &= -2.4\text{V} \end{aligned}$$

Nodal Analysis with Voltage Sources (I)

- How would voltage sources affect nodal analysis?
  - The current through the voltage source is unknown and cannot be expressed with node voltages!
- Depending on what nodes the source is connected to, the approach varies
- Case 1: the V-source connects to the reference node → The voltage at the other terminal of the source is specified → No equation is needed
- Case 2: The V-source connects to two non-reference nodes. Treat the two nodes as an "supernode (超节点)" and apply KCL for the supernode
- A supernode is formed by enclosing a (dependent or independent) voltage source connected between two non-reference nodes and any elements connected in parallel with it → So nodes 2 and 3 form a supernode

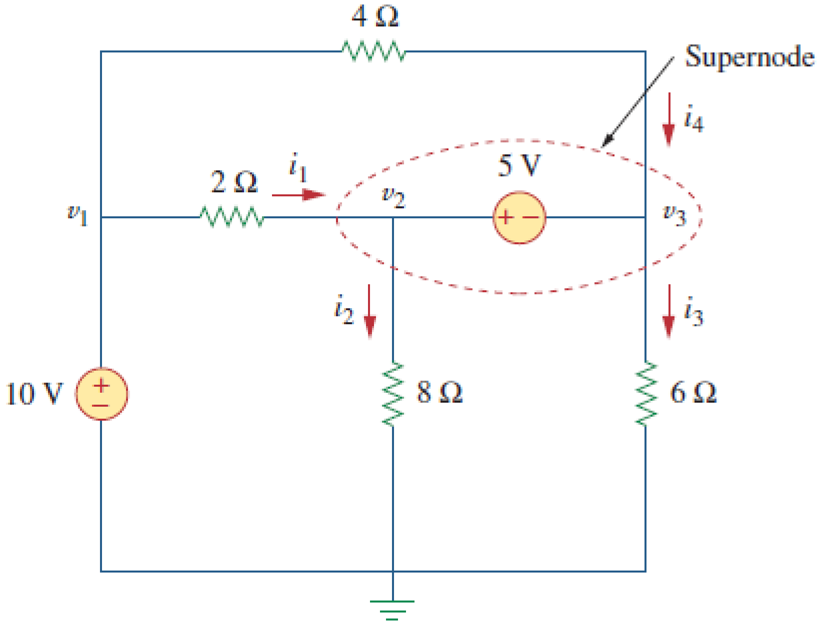


Figure 3.7
A circuit with a supernode.

Nodal Analysis with Voltage Sources (II)

- Why supernode?
  - Nodal analysis requires applying KCL
  - The current through the voltage source cannot be known in advance (Ohm's law does not apply)
  - By lumping the nodes together, the current balance can still be described. The current balance in the supernode of Fig. 3.7 would be

$$i_1 + i_4 = i_2 + i_3$$

- Recall that KCL also applies to a closed boundary → A generalized case → The total current entering the closed surface is equal to the total current leaving the surface

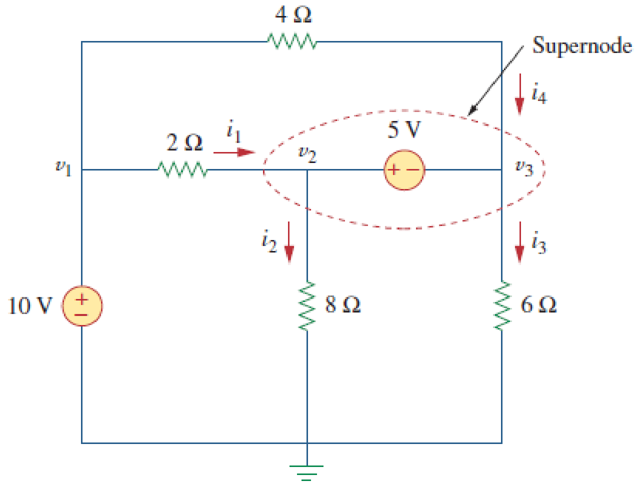


Figure 3.7
A circuit with a supernode.

Nodal Analysis with Voltage Sources (III)

- The 10V source is of 'case 1', and therefore $v_1 = 10\text{V}$
- Apply KCL at the supernode of 5 V source, yield

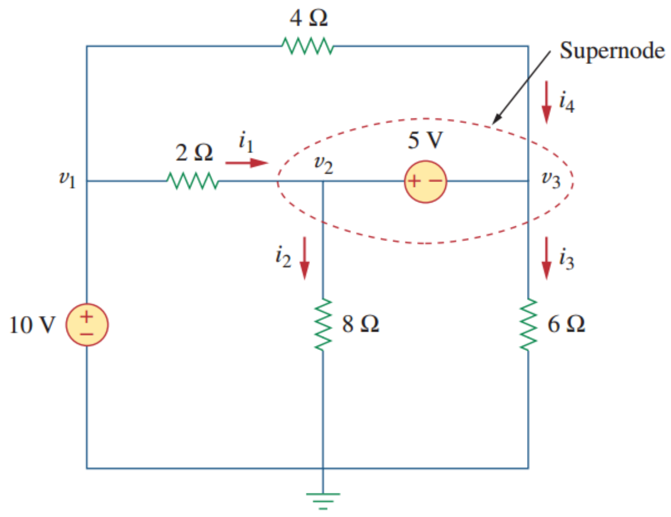
$$i_1 + i_4 = i_2 + i_3$$

$$\frac{v_1 - v_2}{2} + \frac{v_1 - v_3}{4} = \frac{v_2 - 0}{8} + \frac{v_3 - 0}{6}$$

- We still need one equation. It can be obtained by applying KVL to a mesh with the 5 V source as branch (see Fig. 3.8)

$$-v_2 + 5 + v_3 = 0 \quad \Rightarrow \quad v_2 - v_3 = 5$$

- Together with v_1 = 10 V, we can get v_2 = 9.2 V, v_3 = 4.2 V; and so i_1 = 0.4 A, i_2 = 1.15 A, i_3 = 0.7 A, i_4 = 1.45 A



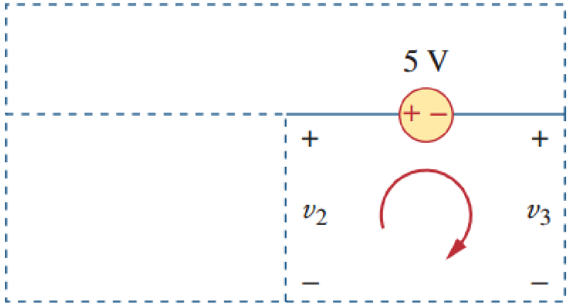
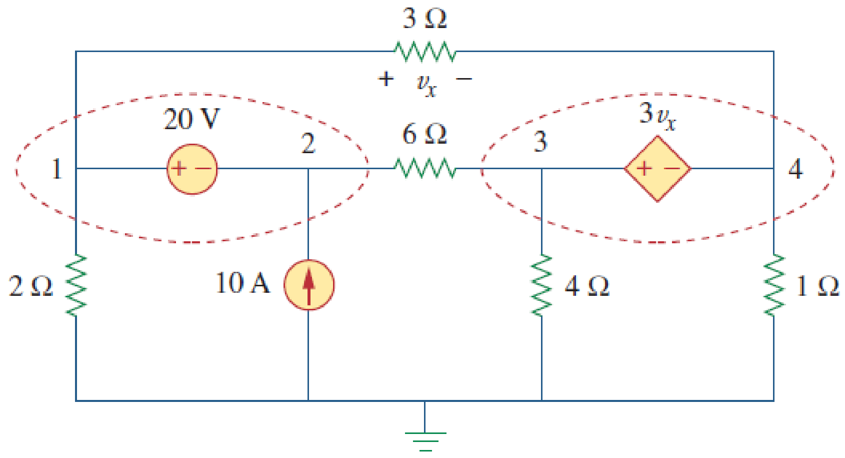


Figure 3.8
Applying KVL to a supernode.

Example: Nodal Analysis with Voltage Sources

Find the node voltages in the circuit of Fig. 3.12. **Example 3.4**



Solution:

Nodes 1 and 2, 3 and 4 form two supernodes. Applying KCL to them, yielding

$$\frac{v_3 - v_2}{6} + 10 = \frac{v_1 - v_4}{3} + \frac{v_1}{2} \qquad \frac{v_1 - v_4}{3} = \frac{v_3 - v_2}{6} + \frac{v_4}{1} + \frac{v_3}{4}$$

Branch equation: $-v_1 + 20 + v_2 = 0 \qquad -v_3 + 3v_x + v_4 = 0$

And $v_x = v_1 - v_4$

Solve the equations, v_1 = 26.67 V, v_2 = 6.667 V, v_3 = 173.33 V, v_4 = -46.67 V

Outline

- Introduction
- Nodal Voltage Analysis (节点电压分析法)
- Nodal Analysis with Voltage Source
- **Mesh Current Analysis (网孔电流分析法)**
- **Mesh Analysis with Current Source**
- **Nodal Versus Mesh Analysis**
- DC Transistors Circuits

Mesh Current Analysis (网孔电流分析法)

- Another general procedure for analyzing circuits is to use the mesh currents instead of element currents as the circuit variables → Reducing the number of equations that must be solved simultaneously
- Mesh analysis applies KVL to find unknown currents
  - Recall: A loop is a closed path with no node passed more than once
  - A mesh (网孔) is a loop that does not contain any other loop within it

Steps:

1. Label the mesh currents as i_1, i_2, …, i_n
2. Apply KVL to each of the n meshes. Write the voltages across each elements in terms of the mesh currents according to the Ohm's law
3. Solve for the mesh currents from the resulting n simultaneous equations

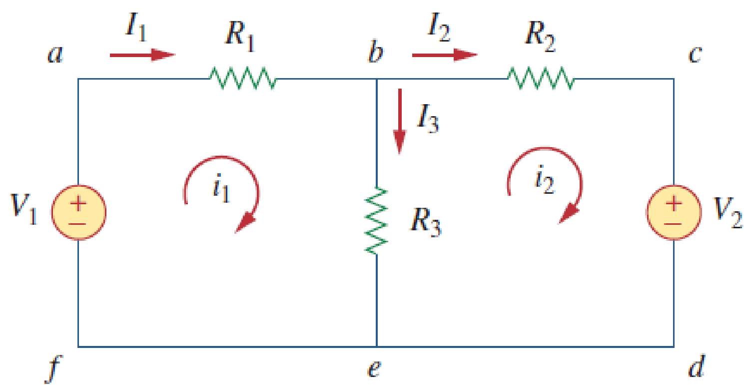


Figure 3.17
A circuit with two meshes.

Mesh Analysis Method (II)

- The first step requires that mesh currents i_1 and i_2 are assigned to meshes 1 and 2
- Although a mesh current may be assigned to each mesh in an arbitrary direction, it is conventional to assume that each mesh current flows clockwise (顺时针)
- As the second step, we apply KVL to each mesh
  - Applying KVL to mesh 1, we obtain

$$-V_1 + R_1 i_1 + R_3(i_1 - i_2) = 0 \quad \text{Or} \quad (R_1 + R_3)i_1 - R_3 i_2 = V_1$$

  - For mesh 2, applying KVL gives

$$R_2 i_2 + V_2 + R_3(i_2 - i_1) = 0 \quad \text{Or} \quad -R_3 i_1 + (R_2 + R_3)i_2 = -V_2$$

- The third step is to solve for the mesh currents, using substitution method, elimination method, Cramer's rule…

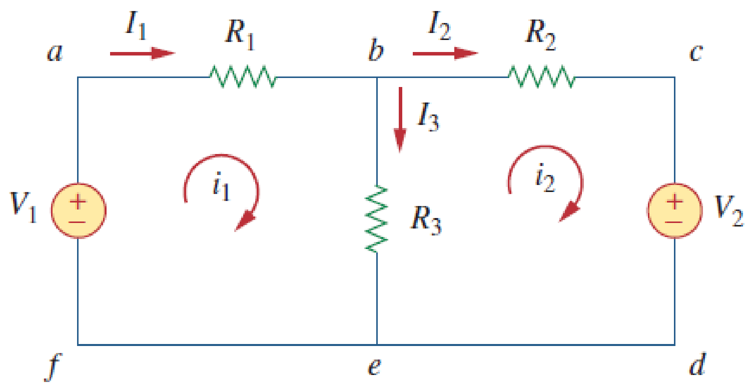


Figure 3.17
A circuit with two meshes.

Example: Mesh Analysis

Example 3.6 Use mesh analysis to find the current I_o in the circuit of Fig. 3.20.

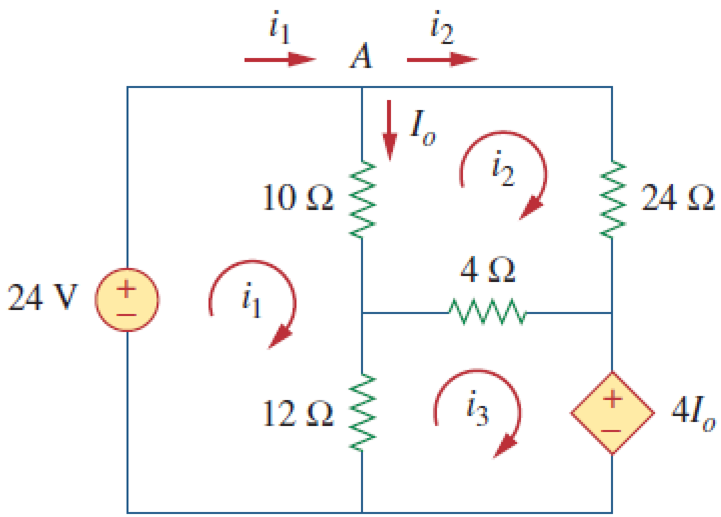


Figure 3.20
For Example 3.6.

- The first step requires that mesh currents i_1 , i_2, and i_3 are assigned to meshes 1, 2, and 3
- Appling KVL to the meshes 1, 2, and 3, yielding

$$\begin{cases} -24 + 10(i_1 - i_2) + 12(i_1 - i_3) = 0 \\ 24i_2 + 4(i_2 - i_3) + 10(i_2 - i_1) = 0 \\ 4I_0 + 12(i_3 - i_1) + 4(i_3 - i_2) = 0 \end{cases}$$

- With KCL at node A

$$I_o = i_1 - i_2$$

- Simplifying and solving the equations, yielding i_1 = 2.25 A, i_2 = 0.75 A, i_3 = 1.5 A. Therefore, $I_0 = i_1 - i_2$ = 1.5 A.

Mesh Analysis with Current Sources (I)

- Case 1: If the current source is located on only one mesh, the current for that mesh is defined by the source → Makes the mesh analysis simpler
- Case 2: If the current source is shared by two meshes → Difficulty to write the voltage across the current source
  - ➢ We create a supermesh (超网孔) by cutting off the current source branch

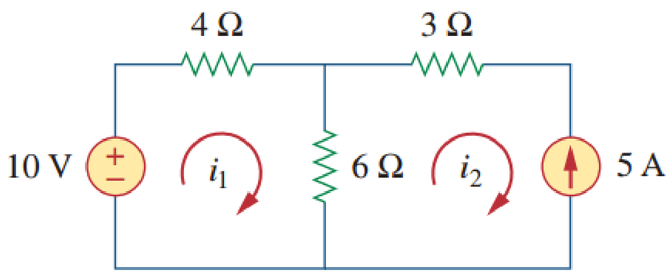


Figure 3.22
A circuit with a current source.

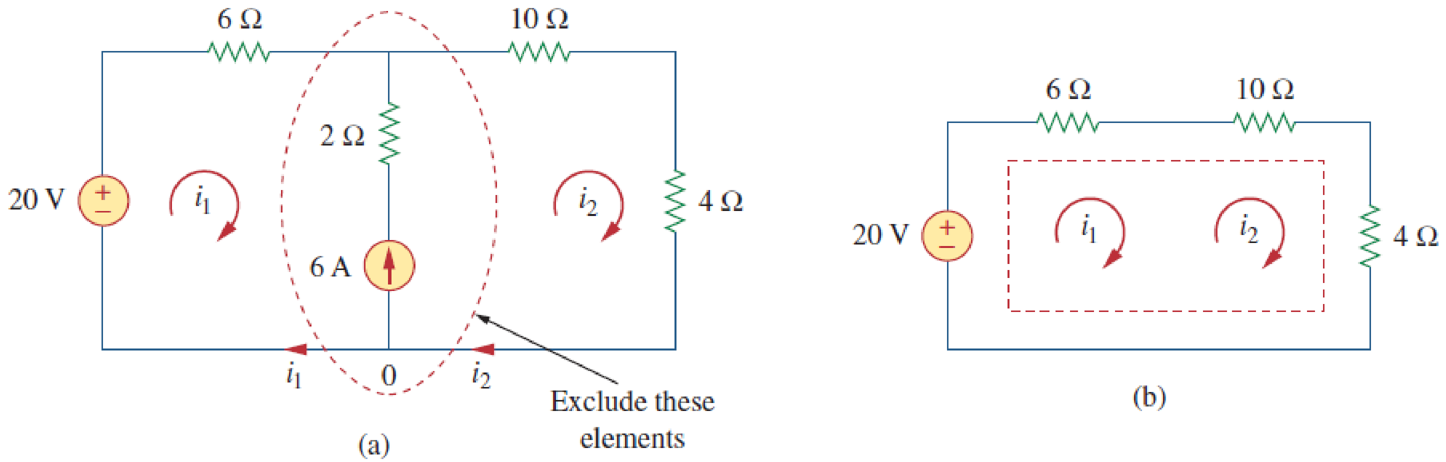


Figure 3.23
(a) Two meshes having a current source in common, (b) a supermesh, created by excluding the current source.

Mesh Analysis with Current Sources (II)

- The two meshes must be joined together, resulting in a supermesh. The supermesh is constructed by merging the two meshes and excluding the shared source and any elements in series with it
- In this example, the 6 A current source is shared between mesh 1 and 2. The supermesh is formed by merging the two meshes. (The current source and the 2 Ω resistor in series with it are removed)
  - Applying KVL to the supermesh, we get

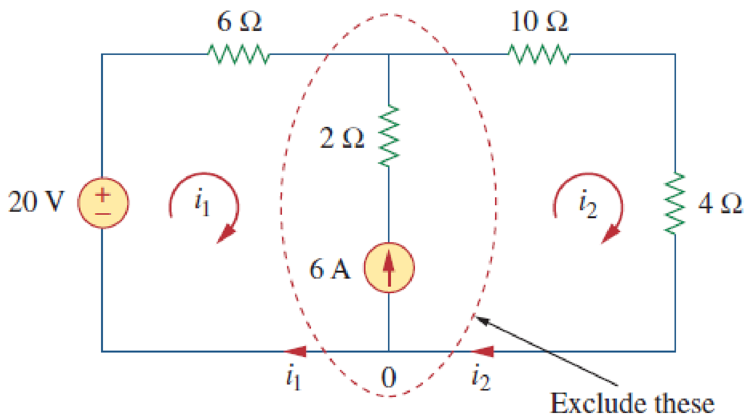
$$-20 + 6i_1 + 10i_2 + 4i_2 = 0$$

  - While at node 0, KCL leads to

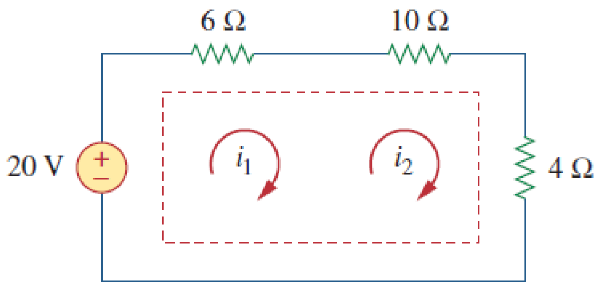
$$i_2 = i_1 + 6$$

  - These give rise to the result of

$$i_1 = -3.2\text{ A}, \qquad i_2 = 2.8\text{ A}$$

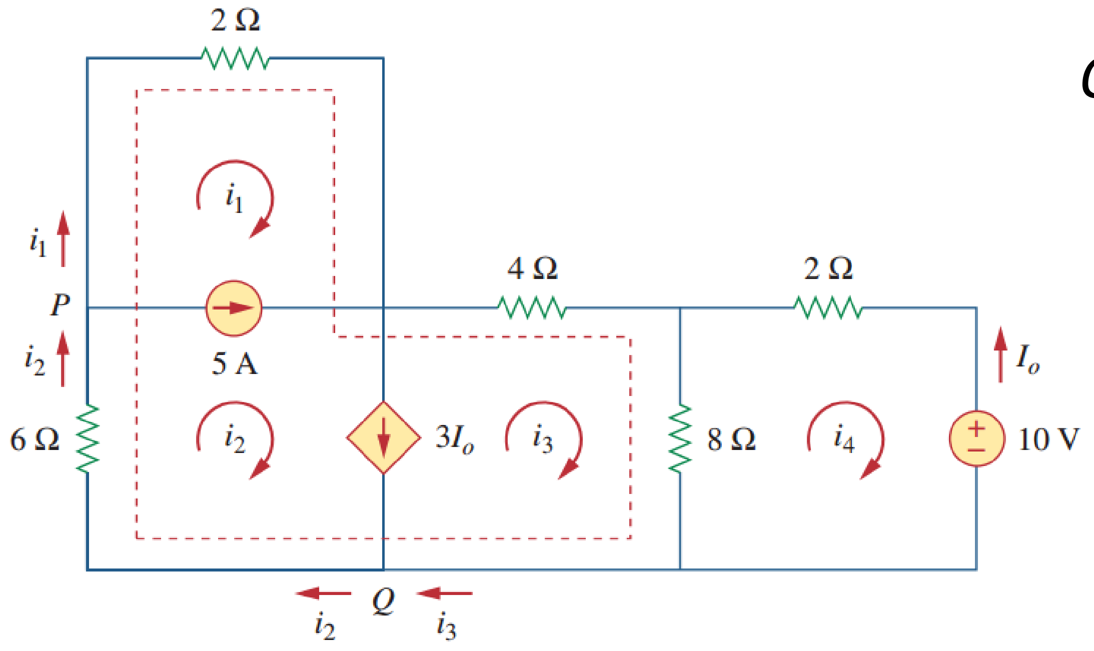


(a)



(b)

Example: Mesh Analysis with Current Sources



Example 3.7

Q: Find i_1 to i_4 in the figure

Solution:

- meshes 1 and 2 form a supermesh
- meshes 2 and 3 form a supermesh
- The two supermeshes intersect and form a larger supermesh as shown.
- Applying KVL to the larger supermesh

$$2i_1 + 4i_3 + 8(i_3 - i_4) + 6i_2 = 0$$

- Applying KCL to nodes *P* and *Q*

$$i_2 = i_1 + 5 \qquad i_2 = i_3 + 3I_o$$

- Applying KVL to mesh 4: $2i_4 + 8(i_4 - i_3) + 10 = 0$
- Taking account of $I_o = -i_4$
- Yielding

$$i_1 = -7.5 \text{ A}, \qquad i_2 = -2.5 \text{ A}$$
$$i_3 = 3.93 \text{ A}, \qquad i_4 = 2.143 \text{ A}$$

Which Analysis Methods to use?

- In principle both the nodal analysis and mesh analysis are useful for any given circuit. What then determines if one is going to be more efficient for solving a circuit problem?
- There are two factors that dictate the best choice:
  - The nature of the network is the first factor
  - The second factor is the information required
- Nodal analysis if:
  - The network contains many parallel connected elements; The network contains many current sources; The network contains many supernodes; The circuit has fewer nodes than meshes (节点数少于网孔数); Node voltages are what are being solved for
- Mesh analysis if:
  - The network contains many series connected elements; The network contains many voltage sources; The network contains many supermeshes; The circuit has fewer meshes than nodes (网孔数少于节点数); Branch or mesh currents are what is being solved for

Selecting an Appropriate Approach (I)

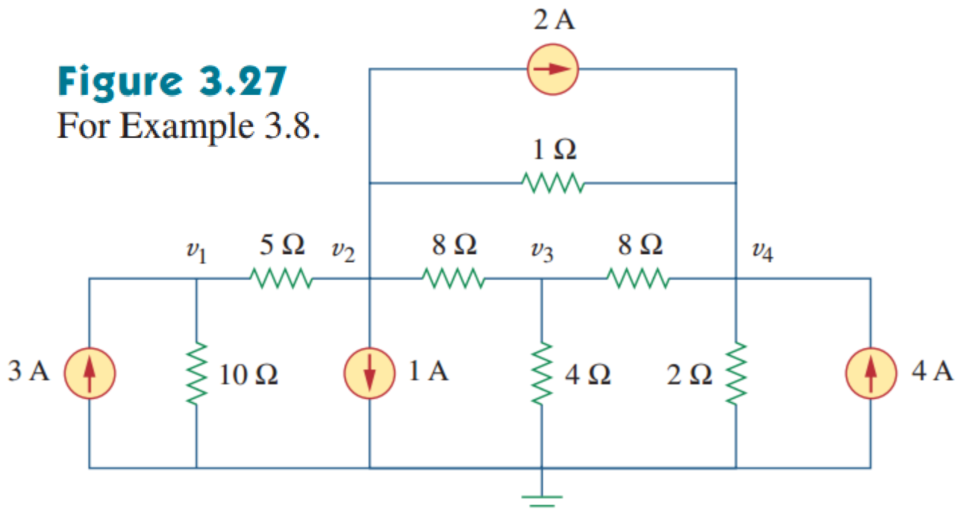


Figure 3.27
For Example 3.8.

Q: Find the node voltages in Fig. 3.27

Solution:

The network contains many parallel connected elements, current sources, has fewer nodes than meshes, and the voltages are what wanted. So, nodal analysis is more suitable

Applying KCL to nodes 1-4, yielding

$v_1/10 + (v_1-v_2)/5 = 3$

$(v_2-v_1)/5 + (v_2-v_4)/1 + (v_2-v_3)/8 = -2 - 1$

$(v_3-v_2)/8 + (v_3-v_4)/8 + v_3/4 = 0$

$(v_4-v_2)/1 + (v_4-v_3)/8 + v_4/2 = 2 + 4$

Simplifying and solving the equations

$$\begin{bmatrix} 0.3 & -0.2 & 0 & 0 \\ -0.2 & 1.325 & -0.125 & -1 \\ 0 & -0.125 & 0.5 & -0.125 \\ 0 & -1 & -0.125 & 1.625 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \\ 0 \\ 6 \end{bmatrix} \Rightarrow \begin{aligned} v_1 &= 13.897\text{V} \\ v_2 &= 5.845\text{V} \\ v_3 &= 3.348\text{V} \\ v_4 &= 7.547\text{V} \end{aligned}$$

Selecting an Appropriate Approach (II)

Q: Find the mesh currents in Fig. 3.29

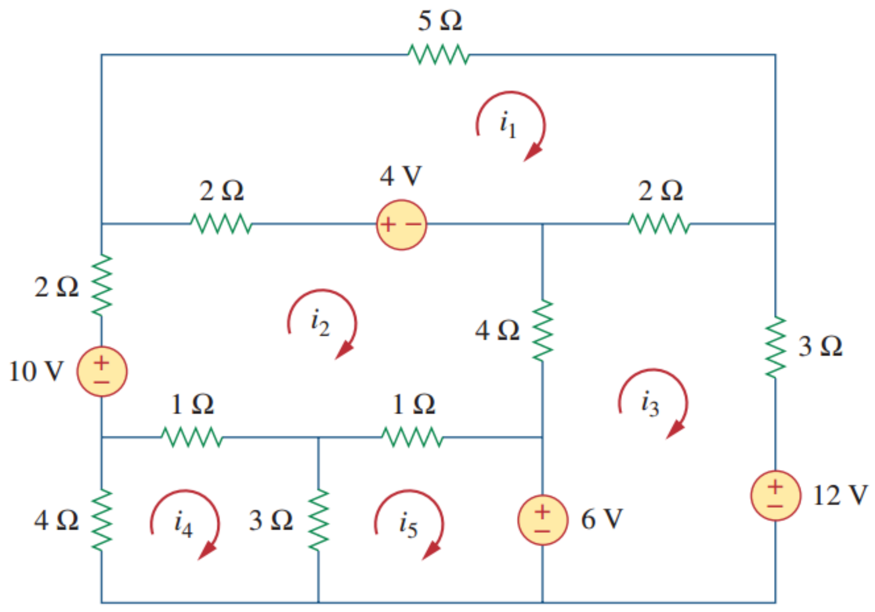


Figure 3.29
For Example 3.9.

Solution:
The network contains many serially connected elements, many voltage sources, fewer meshes than nodes, and the currents are what wanted. So, mesh analysis is selected
Applying KVL to meshes 1-5, yielding

$-4 + 2*(i_1-i_2) + 5*i_1 + 2*(i_1-i_3) = 0$

$-10 + 2*i_2 + 2*(i_2-i_1) + 4 + 4*(i_2-i_3) + 1*(i_2-i_5) + 1*(i_2-i_4) = 0$

$-6 + 4(i_3-i_2) + 2*(i_3-i_1) +3*i_3 + 12 = 0$

$4*i_4 + 1*(i_4-i_2) + 3*(i_4-i_5) = 0$

$3*(i_5-i_4) + 1*(i_5-i_2) + 6 = 0$

Simplifying and solving the equations,

$$\begin{bmatrix} 9 & -2 & -2 & 0 & 0 \\ -2 & 10 & -4 & -1 & -1 \\ -2 & -4 & 9 & 0 & 0 \\ 0 & -1 & 0 & 8 & -3 \\ 0 & -1 & 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ -6 \\ 0 \\ -6 \end{bmatrix} \Rightarrow$$

$I_1 = 0.383$ A; $i_2 = 0.211$ A;
$I_3 = -0.488$ A; $i_4 = -0.718$ A;
$I_5 = -1.986$ A

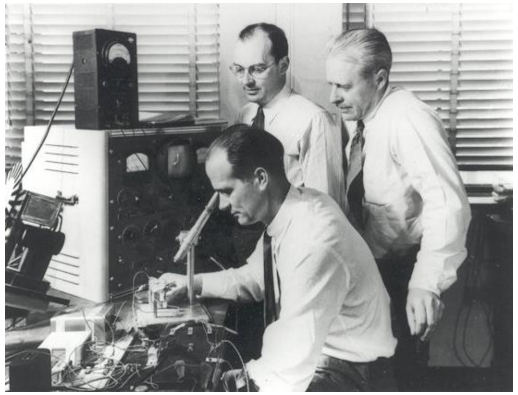
Outline

- Introduction
- Nodal Voltage Analysis (节点电压分析法)
- Nodal Analysis with Voltage Source
- Mesh Current Analysis (网孔电流分析法)
- Mesh Analysis with Current Source
- Nodal Versus Mesh Analysis
- DC Transistors Circuits

Transistor: Introduction

- A transistor (晶体管) is a semiconductor device used to amplify or switch electric signals and electrical power
  - ➢ A basic component for the integrated circuits found in these electronics and computers
- Transistors are divided into two major categories:
  - ➢ Bipolar Junction Transistor (BJT)
  - ➢ Field Effect Transistor (FET)
- Our objective is to present enough detail about FET, especially the most popular type, Metal Oxide Semiconductor Field Effect Transistor (MOSFET, 金属-氧化物半导体场效应晶体管), to enable us to apply the techniques developed in this chapter to analyze dc transistor circuits

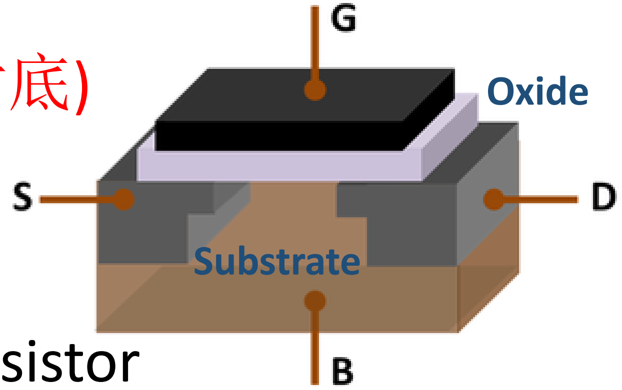
Replica of the first transistor demonstrated in 1947



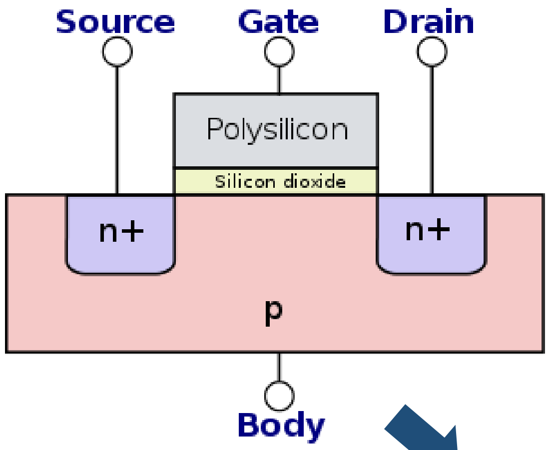
Transistor inventors, John Bardeen, and Walter Brattain, with their supervisor William Shockley (seated) 1948. (Credit: AT&T Archives.)

MOSFET Basics (I)

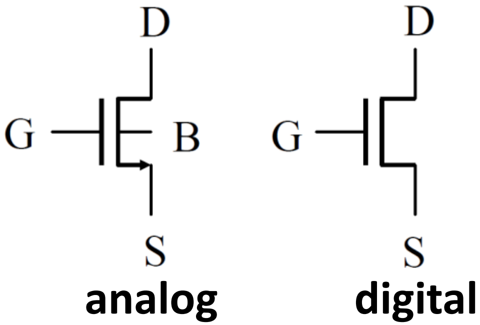
- Four-terminal: Gate (栅极), Drain (漏极), Source (源极), Body (衬底)
- The area between the source and drain is called substrate
- The gate is separated from the substrate by an insulating layer
- There are two types of MOSFET: NMOS transistor and PMOS transistor



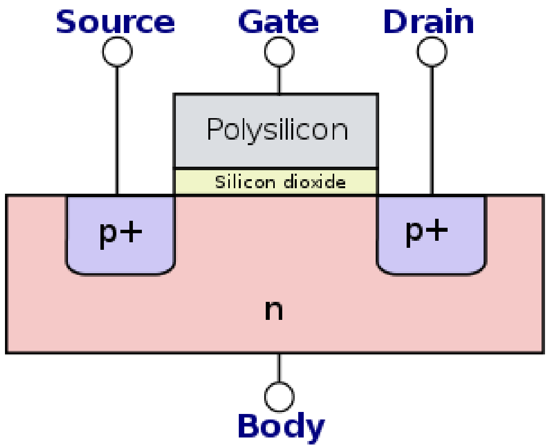
NMOS Structure

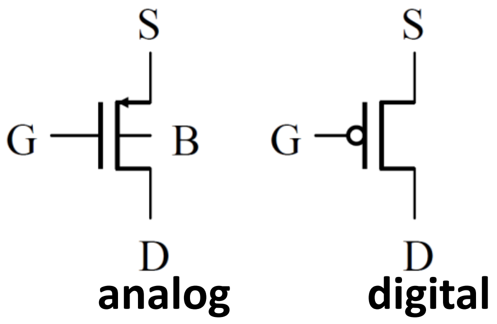


NMOS Symbol

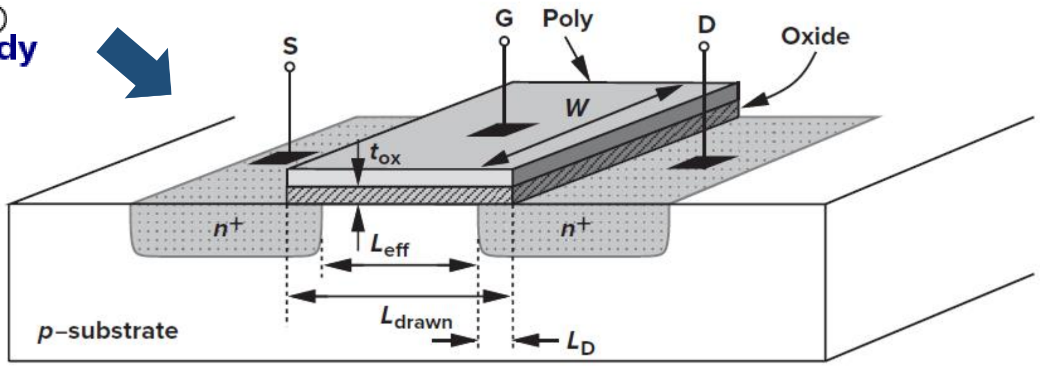


PMOS Structure





PMOS Symbol

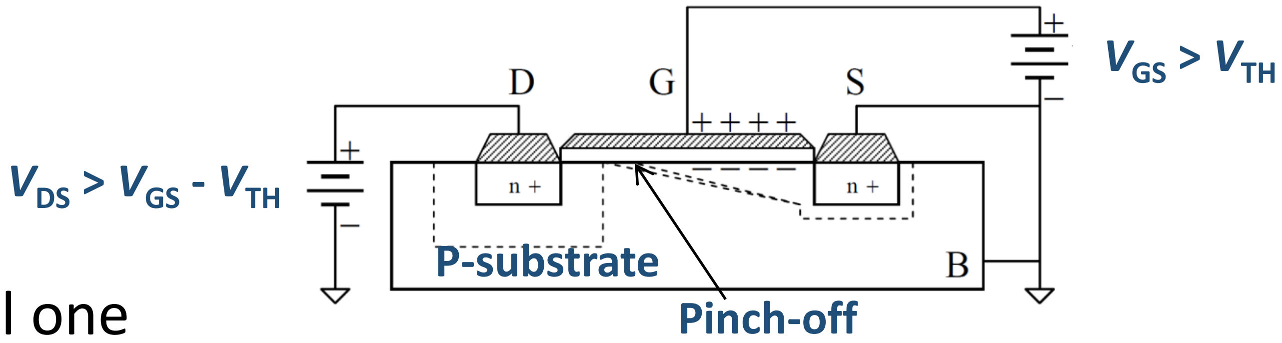


Effective length $L_{eff} = L_{drawn} - 2L_D$

Gate oxide thickness t_{ox}

MOSFET Basics (II)

- Three operating regions:
  - ➢ Cut-Off Region (截止区)
  - ➢ Triode Region (线性区)
  - ➢ Saturation Region (饱和区)



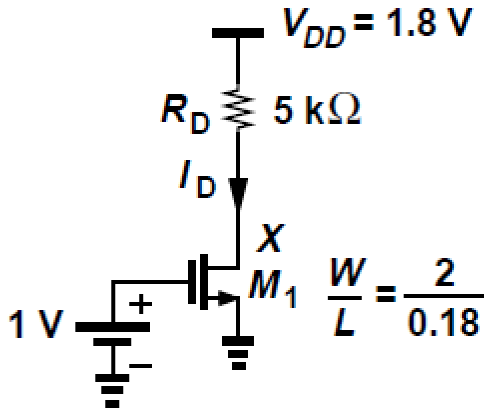
- The saturation region is the most useful one
- **Threshold voltage** (阈值电压, V_{TH}): the minimum gate-to-source voltage V_{GS} that is needed to create a conducting path between the source and drain terminals
- When $V_{GS} > V_{TH}$ and $V_{DS} = V_{GS} - V_{TH}$, the depletion region reduces to zero at the drain (**pinch off**). The current reaches the maximum value, the transistor enters the saturation region (active region)
- The ideally saturate current I_D is given by

$$\boxed{I_D = \frac{\mu_n C_{ox}}{2}\frac{W}{L}(V_{GS} - V_{TH})^2}$$

  - ➢ μ_n is the mobility of electron, C_{ox} is the capacitance per unit area, W is the width of the NMOS, and L is the channel length of the NMOS

Example: Analysis of Circuit Involving Transistor

- Calculate the bias current of M_1 in the following figure. Assume $\mu_n C_{ox}$ = 100 μA/V^2 and V_{TH} = 0.4 V
- What is the drain voltage V_X?



- It is unclear a priori in which region M_1 operates. Let us assume M_1 is saturated and proceed. Since V_{GS} = 1 V

$$\begin{aligned} I_D &= \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2 \\ &= 200\ \mu\text{A}. \end{aligned}$$

- Check our assumption by calculating the drain potential (KVL)

$$\begin{aligned} V_X &= V_{DD} - R_D I_D \\ &= 0.8\ \text{V}. \end{aligned}$$

- The drain voltage is lower than the gate voltage, but by less than V_{TH} → Therefore indicates that M_1 indeed operates in saturation

Summary

- Nodal analysis is the application of Kirchhoff's current law (KCL) at the non-reference nodes. We express the result in terms of the node voltages. Solving the simultaneous equations yields the node voltages
- A supernode consists of two non-reference nodes connected by a (dependent or independent) voltage source
- Mesh analysis is the application of Kirchhoff's voltage law (KVL) around meshes, i.e., loops without other loops inside them. We express the result in terms of mesh currents. Solving the simultaneous equations yields the mesh currents
- A supermesh consists of two meshes that have a (dependent or independent) current source in common
- Nodal analysis is normally used when a circuit has fewer node equations than mesh equations. Mesh analysis is normally used when a circuit has fewer mesh equations than node equations
- DC transistor circuits can be analyzed using the techniques covered in this chapter

- *Acknowledgement: This Lecture Note is partially based on materials from Prof. Chenchang Zan and partially adapted from the notes of the book "Fundamentals of Electric Circuits" by C. K. Alexander and M. N. O. Sadiku*

Cramer's Rule (克拉默法则)

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned} \quad \Rightarrow \quad \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ldots & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

The solution of linear simultaneous equations by Cramer's rule boils down to finding

$$x_k = \frac{\Delta_k}{\Delta}, \qquad k = 1, 2, \ldots, n \qquad \textbf{(A.12)}$$

where Δ is the determinant of matrix **A** and Δ_k is the determinant of the matrix formed by replacing the kth column of **A** by **B**.

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}, \qquad \Delta_1 = \begin{vmatrix} b_1 & a_{12} & \cdots & a_{1n} \\ b_2 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ b_n & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

$$\vdots \qquad\qquad\qquad \vdots$$

$$\Delta_2 = \begin{vmatrix} a_{11} & b_1 & \cdots & a_{1n} \\ a_{21} & b_2 & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & b_n & \cdots & a_{nn} \end{vmatrix}, \ldots, \Delta_n = \begin{vmatrix} a_{11} & a_{12} & \cdots & b_1 \\ a_{21} & a_{22} & \cdots & b_2 \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & b_n \end{vmatrix}$$